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RESEARCH

Adaptive Fusion Attention for enhanced classification and interpretability in medical imaging

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Abstract
Accurate medical image classification is crucial for effective clinical decision support, improving patient outcomes and reducing healthcare costs. However, developing expert Computer-Aided Diagnosis systems for accurate medical image classification remains a challenging task. Recent advancements in attention mechanisms have revolutionized deep learning-based approaches, leading to improved performance even in applications with limited labeled data. Despite these advances, challenges such as overfitting and poor generalization persist. This work introduces a novel deep learning-based model that incorporates Adaptive Fusion Attention to enhance medical image analysis. The proposed attention module employs a hierarchical fusion of spatial and temporal attention mechanisms, complemented by adaptive refinement. This enables the model to focus on the most discriminative features in medical images, improving its ability to detect abnormalities. Additionally, GRAD-CAM visualizations demonstrate that the model effectively highlights pathological regions while minimizing attention on non-relevant areas. The model is evaluated on three benchmark datasets-APTOS-2019, Figshare, and SARS-CoV-2-demonstrating its effectiveness in Diabetic Retinopathy grading, Brain Tumor Classification, and COVID-19 detection. Experimental results show substantial improvements, achieving 84.56% accuracy for retinopathy grading, 99.60% accuracy for tumor classification, and 99.35% accuracy for COVID-19 detection. These results, along with superior performance across other metrics such as ROC-AUC, and F1-scores, demonstrate the effectiveness of the proposed model with Adaptive Fusion Attention over existing approaches.

Keywords Medical image classification · Deep learning · Adaptive Fusion Attention · Transfer learning · Pre-trained models · Diabetic retinopathy (DR) · Brain tumor classification (BTC) · Severe acute respiratory syndrome CoronaVirus 2 (SARS-CoV-2) · Gradient centralization

1 Introduction

The preservation of human life is paramount, and safeguarding health is a critical priority. Various factors such as population growth, lifestyles, dietary changes, and pollution, have significantly impacted public health, contributing to the prevalence of conditions such as diabetes, brain tumors, cardiovascular diseases, and infectious diseases like COVID-19. Diabetes, for instance, disrupts insulin production and can affect organs like the kidneys, heart, and eyes, while brain tumors arise from abnormal cell growth, often due to carcinogens or radiation [2]. Furthermore, viruses like COVID-19 pose severe global health threats, underscoring the necessity of early diagnosis and preventive healthcare. Effective public health measures and access to timely diagnostic tools are vital for mitigating these health challenges [13].

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Despite advances in healthcare, diagnostic services remain limited, particularly in rural regions and among populations below the poverty line. To bridge this gap, Computer Assisted Diagnosis (CAD) systems have become invaluable tools for accurate and timely diagnosis. Recent research has emphasized developing automated techniques that leverage medical imaging to evaluate organ conditions, with deep learning (DL) models demonstrating notable success in these applications [12]. The superior potential of Deep learning in computer vision tasks over traditional methods, makes it indispensable for medical image analysis. However, existing deep learning approaches are often tailored to specific organs or diseases, lacking a versatile, generalized model suitable across various tasks.

Medical image analysis has evolved from classical image processing techniques to sophisticated models utilizing machine learning, Deep Neural Networks (DNNs), transformers, and attention mechanisms [22]. Traditional approaches primarily relied on manual feature extraction, which was limited in scope and inadequate for the complexities inherent to high-dimensional medical imaging data. The advent of machine learning introduced data-driven techniques that improved accuracy in classification and detection tasks, yet these models still faced limitations in capturing subtle but critical spatial and temporal dependencies.

Deep learning has revolutionized the field of medical image analysis, with significant advancements in the automatic classification, detection, and grading of various medical conditions. Automated systems for tasks such as brain tumor classification (BTC), diabetic retinopathy (DR) grading, and COVID-19 diagnosis have demonstrated the potential to support clinical decisions with enhanced speed, accuracy, and reliability. CNNs proved effective by automating feature extraction, achieving high accuracy across various tasks, yet they are limited in the ability to selectively focus on critical regions within medical images. Recent developments in attention mechanisms have further transformed medical image analysis. Attention blocks within Convolutional Neural Networks (CNNs) allow models to prioritize essential spatial and channel-wise information, enhancing interpretability and focusing on disease-relevant regions [66]. However, many conventional attention methods fall short in dynamically adapting to the unique requirements of complex medical images, especially where both spatial and temporal dependencies play a crucial role. However, these models often face limitations in effectively handling the complex, high-dimensional nature of medical image data, which requires an advanced understanding of spatial and temporal dependencies to capture subtle but critical features accurately [34].

To address these challenges, our work introduces a novel Adaptive Fusion Attention (AFA) mechanism that enhances the interpretability and adaptability of medical imaging models. The AFA mechanism captures both spatial and channel-wise dependencies, dynamically refining its focus based on the unique requirements of each image, thereby improving classification and diagnostic accuracy. The proposed AFA mechanism leverages advanced attention capabilities, incorporating hierarchical fusion and adaptive gating, which allows the model to dynamically adjust its focus to enhance learning representations. The proposed model incorporates this AFA block to create a more adaptable and context-sensitive attention mechanism, specifically tailored for the demands of medical imaging. The AFA block provides hierarchical fusion, integrating spatial and channel attention, which enables it to capture dependencies that standard attention mechanisms often miss. This mechanism is further refined with adaptive convolution layers, which dynamically adjust attention weights, allowing the model to focus on the most critical features in complex medical images.

To rigorously evaluate the proposed AFA mechanism, we selected three distinct and critical medical imaging tasks: brain tumor classification (BTC), diabetic retinopathy (APTOS2019), and SARS-CoV-2 (COVID-19) diagnosis. These tasks were chosen to demonstrate versatility of the model across various types of medical imaging challenges:

- *Brain Tumor Classification (BTC)* provides a test case for AFA's ability to enhance interpretability by isolating critical regions in complex, high-resolution images. With high mortality rates associated with brain tumors, early detection and accurate classification are essential for effective treatment.
- *Recognizing Diabetic Retinopathy (DR)* is an ideal application for evaluating the AFA mechanism's spatial precision. DR detection requires distinguishing subtle features that signify disease progression, crucial for preventing vision loss in diabetic patients.
- *SARS-CoV-2 Diagnosis (COVID-19)* highlights the importance of robust, adaptable diagnostic support during a pandemic. The AFA's capacity to handle varying visual patterns is tested here, as COVID-19 diagnosis involves distinguishing pulmonary conditions with overlapping visual characteristics.

By selecting BTC, DR, and Covid-19, we aim to demonstrate the AFA mechanism's adaptability, robustness, and clinical applicability across diverse real-world diagnostic settings.

The primary contributions of this work are summarized as follows:

- Development of a AFA mechanism that integrates spatial and channel-wise dependencies for effective medical image representation learning
- Introduction of an adaptive attention refinement to dynamically adjust focus based on input-specific characteristics
- Achievement of high performance across diverse benchmark datasets, demonstrates the robustness of the proposed model
- Enhanced interpretability by highlighting disease-relevant image regions, facilitating more informed clinical decisions
- Establishment of new benchmarks in attention-based medical imaging applications

2 Background study

Medical imaging plays an indispensable role in modern healthcare, enabling early detection and accurate diagnosis of various medical conditions. With the advent of deep learning techniques, especially convolutional neural networks (CNNs), the field of medical image analysis has propelled advancements. These advancements have facilitated the automation of image classification and disease detection, providing clinicians with essential tools to enhance patient care. While CNNs excel at extracting features and learning patterns directly from images, challenges remain in the interpretability of deep learning models, particularly in understanding how these models arrive at specific decisions. This literature review explores the evolution of medical imaging technologies, deep learning application in this field, and the incorporation of attention mechanisms, framing the context for the proposed AFA mechanism.

Over recent decades, medical imaging modalities such as magnetic resonance imaging (MRI), fundus photography, and chest computed tomography (CT) scans have become indispensable diagnostic tools for a range of health conditions, including brain tumors, diabetic retinopathy (DR), and COVID-19. MRI is essential for brain tumor classification (BTC), providing high-resolution images of brain structures and enabling precise localization of tumor boundaries, crucial for early diagnosis and treatment [11]. Similarly, fundus photography offers a non-invasive approach to detecting DR, allowing clinicians to observe subtle retinal changes associated with diabetes [25]. For infectious diseases like COVID-19, chest CT scans reveal characteristic lung patterns, such as ground-glass opacities, that are indicative of infection severity and progression [61]. These modalities, while effective, require significant interpretative expertise, underscoring the need for automated analysis systems to support clinicians in decision-making [58].

Deep learning has revolutionized medical image analysis by automating feature extraction and classification processes. Convolutional neural networks (CNNs) have been particularly effective, achieving state-of-the-art results in various medical imaging tasks. For instance, the U-Net architecture has gained prominence for its effectiveness in biomedical image segmentation, significantly improving the delineation of tumor boundaries in MRI scans [4]. Additionally, several studies have demonstrated the application of CNNs in brain tumor classification, where models have been trained to distinguish between benign and malignant tumors with high accuracy [37]. Similarly, deep learning models have shown promise in diabetic retinopathy detection, with algorithms achieving accuracies comparable to those of expert ophthalmologists [2]. Convolutional autoencoders focus on deep feature learning to enhance the representation of medical images, enabling better performance in various diagnostic tasks and leading to improved classification and segmentation outcomes [22].

Despite the successes of CNNs in medical imaging tasks, their inability to focus on critical image regions selectively limits their effectiveness. This is particularly challenging in applications that require a detailed understanding of subtle spatial features, such as distinguishing tumor boundaries or grading DR severity. Traditional CNNs lack the capacity to selectively prioritize disease-relevant areas, which is essential in high-stakes diagnostics where model transparency and reliability are paramount [59]. This limitation has driven the development of attention mechanisms as a means to improve both interpretability and performance in medical imaging tasks [17].

Attention mechanisms have emerged as a solution to the limitations of CNNs, enabling models to focus selectively on crucial image features. Self-attention and spatial attention techniques allow models to learn dependencies within an image, prioritizing regions that hold significant information about the disease [67]. Recent studies have integrated attention mechanisms into CNN architectures, leading to improved classification results and interpretability in medical image analysis [59]. For example, channel attention modules like the SE (Squeeze-and-Excitation) block and CBAM (Convolutional Block Attention Module) improve interpretability by highlighting disease-relevant regions [66]. However, these mechanisms often focus on either spatial or channel-wise information in isolation, which may not suffice for complex medical images that require both spatial and channel dependencies [64].

The introduction of transformer-based architectures has further advanced the capabilities of medical image analysis. Transformers leverage self-attention mechanisms, allowing for better handling of region dependencies within images. Transformer models have been successfully applied to

applications like COVID-19 diagnosis from chest X-rays, where they outperform CNNs by leveraging contextual information across the entire image [34]. These architectures are particularly beneficial in tasks like BTC, where long-range dependencies and spatial attention are critical for detecting tumor spread [44]. However, transformers face challenges in interpretability, essential for understanding model decisions in medical contexts. Transformers struggle with interpretability, which is crucial for understanding model decisions in medical contexts. Their high computational demands hinder use in resource-limited settings, and they often need large, diverse datasets that are scarce in medical imaging. Additionally, transformers may lack robustness on unseen data, increasing the risk of overfitting to the training distribution [33]

Although attention mechanisms have improved the interpretability of deep learning models in medical imaging, current approaches face challenges when applied to diverse tasks requiring spatial and temporal awareness. Most models focus on a specific task or disease, lacking adaptability across different medical applications. Moreover, existing attention mechanisms tend to handle spatial and channel dependencies separately, limiting their capacity to capture complex interdependencies required for nuanced tasks like brain tumor classification or diabetic retinopathy severity grading [64].

Medical imaging applications highlight the necessity for advanced attention mechanisms to address specific challenges:

2.1 Literature on diabetic retinopathy recognition

Aiming at the early detection of diabetic retinopathy to prevent irreversible damage to vision, spurred significant research in the development of automated DR diagnostic systems. Initially, traditional machine learning methods, such as Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN), formed the cornerstone of DR detection systems [63]. These methods relied heavily on hand-crafted features such as Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG), which were effective but had limitations in capturing the intricate patterns and subtle variations in medical images, requiring substantial domain expertise for feature engineering [48]. These approaches also struggle with handling the complexity and variability in retinal images, often failing to generalize well across different patient populations and imaging conditions [28].

With the rise of deep learning (DL) technologies, DR recognition experienced a paradigm shift. Convolutional Neural Networks (CNNs) emerged as the dominant architecture for automating feature extraction directly from raw image data. The ability of CNNs to learn hierarchical representations of features from images allowed for more accurate and efficient DR detection [5]. Notably, architectures like U-Net, originally developed for medical image segmentation, gained prominence in DR detection due to their strong performance in segmenting the retinal fundus images and isolating key features associated with DR, such as microaneurysms, exudates, and hemorrhages. These networks offered a significant improvement in delineating structures critical for the classification of DR severity [38].

A major leap forward in improving model accuracy and interpretability came with the incorporation of attention mechanisms into CNN architectures. Attention mechanisms allow models to focus on the most relevant areas of an image-such as the central retinal regions where diabetic lesions are likely to occur-while disregarding less relevant background noise [54]. This ability to enhance spatial focus not only improves classification accuracy but also makes models more interpretable, providing visual explanations for their predictions [38]. The use of attention has become increasingly prevalent, with models like Lesion Aware Attention Networks and Hinge Attention Networks specifically designed to enhance the attention focus on regions most indicative of DR [53].

The availability of large-scale datasets, such as the Messidor database and the APTOS 2019 dataset, has been pivotal in advancing these deep learning models. These rich datasets allow the training of models on diverse and comprehensive retinal images, facilitating the generalization of models across different patient demographics, geographic regions, and imaging conditions [16]. However, challenges persist in dealing with the inherent variability in retinal image quality, imaging conditions, and the complex nature of DR pathology, which often necessitate model adaptability to perform well across different settings [21].

In recent years, research has shifted towards more sophisticated architectures that leverage the power of transformers. These models, which have been proven successful in sequential data tasks such as natural language processing, are gaining traction in medical image analysis due to their ability to capture long-range dependencies within the images [58]. Transformer-based models, such as MedTransNet have shown considerable promise in DR recognition, as they can better handle complex patterns and relationships within retinal images compared to conventional CNNs [15]. These models not only improve accuracy but also provide a mechanism to capture global context, which is crucial in identifying lesions that may be spread over larger areas of the retina.

The continuous evolution of DR detection models has been driven by both the availability of new data and the development of increasingly sophisticated deep learning techniques. From traditional machine learning methods to

the latest transformer-based models, each step has brought DR recognition closer to achieving reliable and accurate automated diagnoses. As these models become more capable, they will play an increasingly important role in the early detection of DR, providing significant improvements in patient outcomes and reducing the burden on healthcare systems worldwide.

2.2 Literature on brain tumor classification

In brain tumor classification, traditional image processing techniques were initially prominent, relying on machine learning models that extracted hand-crafted features from magnetic resonance imaging (MRI) scans. Early methods utilized descriptors such as the Gray Level Co-occurrence Matrix (GLCM) and Speeded-Up Robust Features (SURF), combined with machine learning classifiers for tumor identification [43]. These methods provided a foundational approach but were limited by their dependence on manually extracted features, which often lacked the capacity to capture the complex patterns in MRI data effectively.

With the advent of deep learning, convolutional neural networks (CNNs) transformed the field by allowing for automatic feature extraction, which significantly improved classification performance [16]. CNN architectures like VGG16 and DenseNet emerged as influential models, demonstrating substantial accuracy in differentiating between benign and malignant tumors due to their deep hierarchical structures, which can capture intricate spatial patterns in MRI images [8].

Further advances incorporated attention mechanisms within CNNs to enhance model focus on critical regions of MRI scans, thus improving the accuracy and robustness of tumor classification [62]. For instance, approaches like TumorAwareNet utilize attention modules to prioritize tumor regions, optimizing the model ability to distinguish between subtle tissue differences [17]. Attention mechanisms marked a pivotal shift, as they allowed models to better emulate clinical focus areas, yielding more accurate and interpretable results [16].

As deep learning continued to evolve, researchers began exploring transformer models for medical imaging, particularly their ability to capture long-range dependencies and contextual relationships within images. Transformers, initially popularized in natural language processing, have shown promise in handling the spatial hierarchies and dependencies present in brain tumor classification tasks, adding a new dimension to model accuracy and interpretability [36]. Despite these advancements, challenges such as data scarcity and overfitting remain. To address this, researchers have increasingly turned to transfer learning techniques, adapting pre-trained models to brain tumor datasets [10]. Transfer learning leverages knowledge from large datasets, enabling robust performance even with limited labeled data [68]. This approach not only reduces the need for extensive data but also improves generalization, making it valuable for practical medical applications. Moreover, interpretability has become a focus in brain tumor classification, as researchers aim to develop models that offer insights into their decision-making processes. This need for transparency in medical AI has spurred investigations into interpretable models, balancing high accuracy with the ability to provide explanations for predictions [51].

In parallel, novel architectures have emerged, such as Capsule Networks, which were designed to overcome CNNs' limitations in capturing spatial hierarchies. While Capsule Networks showed promise, they struggled with accuracy due to the complexity of brain tumor images [4]. Hybrid models combining CNN and deep neural networks (DNN) also saw early application; these models used CNNs for feature extraction and DNNs for classification but were eventually outpaced by deeper and more sophisticated neural networks [43]. Fine-tuning pre-trained models like VGG-19 further enhanced accuracy by adjusting the model specifically for brain tumor images. However, VGG-19 lacks an attention mechanism, limiting its focus on critical regions and affecting its performance relative to attention-based models [60]. Following this, dual-channel DNN setups emerged, where different data aspects were processed separately to improve classification accuracy, though still falling short of newer models with more complex architectures [19].

Models combining AlexNet with SVM classifiers introduced robustness through support vector machine-based generalization, but this combination was ultimately limited by the SVM's inability to capture subtle image features effectively, a task that more advanced neural networks could manage better [49]. Custom CNN + SVM setups provided similar improvements, though they too could not match the complexity required to handle nuanced features in MRI images [24].

Using transfer learning with VGG-19, researchers attained enhanced metrics such as precision, recall, F1-score, and ROC AUC, leveraging pre-trained VGG-19 weights for performance gains, albeit without attention mechanisms [1]. Building on this, 3D-CNN models introduced 3D convolutions to capture volumetric features, proving effective in medical image classification but requiring substantial computational resources [26].

The combination of CNNs and RNNs in hybrid CNN-RNN models marked another significant step, with RNNs capturing sequential dependencies within MRI slices, further enhancing classification accuracy [35]. Multi-modal Squeeze and Excitation Networks (MSENet) utilized a

different approach, focusing on highlighting critical regions in brain tumor images through selective attention, demonstrating significant accuracy improvements [20].

Subsequent work explored Multi-Level Attention Networks, where two levels of attention were applied to improve focus on relevant MRI regions. This hierarchical attention mechanism offered a competitive approach in brain tumor classification [55]. More recently, MedTransNet introduced transformers for handling long-range dependencies within images. Although MedTransNet brought innovations with its transformer architecture, it lacked certain specific attention capabilities compared to models like MSENet [58].

In summary, brain tumor classification has evolved from traditional image processing techniques to sophisticated deep learning architectures, incorporating CNNs, RNNs, attention mechanisms, transfer learning, and transformers. Each development has progressively improved model performance, leading to more accurate and interpretable classification systems that hold promise for real-world clinical applications.

2.3 Literature on Covid-19 prediction

In the context of the COVID-19 pandemic, medical imaging has become an essential tool for diagnosis and monitoring disease progression. Initially, reverse transcription-polymerase chain reaction (RT-PCR) testing served as the primary diagnostic method; however, the demand for rapid testing prompted the exploration of imaging modalities like chest X-rays and CT scans [69]. Literature reveals that deep learning techniques, particularly CNNs, have achieved remarkable accuracy in analyzing chest CT scans and X-rays, effectively identifying COVID-19 infections [47]. Models such as SqueezeNet and ResNet are frequently cited for their performance in detecting the disease from imaging data [50].

Transfer learning has been a prevalent approach in the context of COVID-19 diagnosis, allowing for the adaptation of pre-trained models on extensive datasets to improve performance on COVID-19-specific datasets [39]. The application of attention mechanisms in these models enhances their ability to prioritize critical features associated with COVID-19 infection [46]. However, challenges related to computational demands and the need for large labeled datasets persist. Future research is directed toward developing more efficient models capable of generalizing across diverse populations, improving the overall diagnostic landscape for COVID-19 and potentially other respiratory illnesses [45].

Overall, the integration of machine learning, deep learning, attention mechanisms, and emerging transformer architectures continues to reshape the field of medical imaging. These technologies are facilitating advancements in diagnostic accuracy, efficiency, and interpretability, paving the way for more effective healthcare solutions.

Most of the discussed approaches have demonstrated the effectiveness of Transfer Learning in overcoming the limited annotated data problem for training deep neural network architectures. However, capturing micro lesion features and emphasizing category-specific information in medical images remains a significant challenge. In recent years, various visual attention mechanisms have been proposed in the literature to address this issue and capture category-specific micro lesion information from diverse medical imaging modalities [57].

For the task of retinopathy severity classification, a Composite Deep Neural Network with a gated attention mechanism [18] and a Multi-scale Attention network [6] have been proposed. To mitigate the curse of dimensionality resulting from high-dimensional attention feature vectors, a stacked Convolutional Auto-encoder based Spatial Attention model has been introduced [15]. Additionally, a Lesion-aware attention network with Neural Support Vector Machine has been developed to ensure generalizability in classifying retinopathy grades [53]. In contrast to these models, a Hinge Attention Network (HA-Net), which combines Convolutional Auto-encoder and Attention-based Classifier, has been trained in an end-to-end manner to learn attention-based features and predict retinopathy severity labels [54].

In the domain of brain tumor classification, a Multi-modal Squeeze and Excitation Network has been designed to incorporate features from two different pre-trained neural networks and accurately diagnose different types of brain tumors [20]. Furthermore, a Multi-level Attention Network (MA-Net) has been proposed to pay attention to features at multiple levels, enabling precise identification of brain tumor types and severity [55].

Although these attention-based models have demonstrated their robustness in their respective tasks, they still face challenges regarding inter-class similarity (*the degree of resemblance between different classes or categories in a dataset*) and intra-class dissimilarity (*the degree of measure in the variation or differences within individual classes or categories in the dataset*) across different medical imaging modalities, as they were specifically designed for single applications. Moreover, there is still room for improvement in the accuracy of systems proposed in the literature for tasks such as Diabetic Retinopathy Severity Diagnosis, Multi-class Brain Tumor Classification, and COVID-19 Diagnosis. To address these limitations, this work aims to design a generalized Deep Learning approach for accurate classification of medical images with varied modalities, leading to the development of expert computer-aided diagnosis systems. To achieve this, an Advanced Gating

Transformer Network has been designed and implemented for precise medical image diagnosis.

3 Adaptive Fusion Attention Network (AFANet)

Deep Convolutional Network with the proposed methodology aims to optimize medical image analysis through an advanced attention mechanism called AFA. This attention mechanism dynamically adjusts its focus based on image-specific contextual information, enhancing ability of the model to detect critical patterns. Unlike traditional attention approaches, which often treat spatial and channel-wise attention separately, AFA unifies both spatial and channel attentions within a single framework and makes especially suitable for applications requiring fine-grained focus on dynamic contexts, such as medical image analysis. Figure 1 illustrates the architecture of the proposed Adaptive Fusion Attention Network (AFANet), which integrates Adaptive Fusion Attention to improve the accuracy and interpretability of disease prediction models.

The following sections outline each block of the Adaptive Fusion Attention Network in detail, highlighting the technical advancements of the proposed methodology.

3.1 Preprocessing

In the preprocessing phase, medical images undergo essential transformations to prepare them for training, ensuring they are standardized and consistent.

Initially, all images are resized to a uniform dimension of 312 × 312 pixels, required by the pre-trained convolutional neural network used in subsequent stages. For resizing of the images, the cubic interpolation method, specifically the inter-cubic technique is employed. This technique provides smooth and accurate results during both upscaling and downscaling by using 4 × 4 neighborhood of pixels (16 pixels in total) to compute new pixel values with cubic polynomials. This approach achieves high-quality interpolation suitable for medical images that demand sharpness and detail.

Following resizing, images are normalized to a range between 0 and 1. This scaling ensures consistency across input data, enhancing convergence during neural network training by maintaining stable gradients and improving

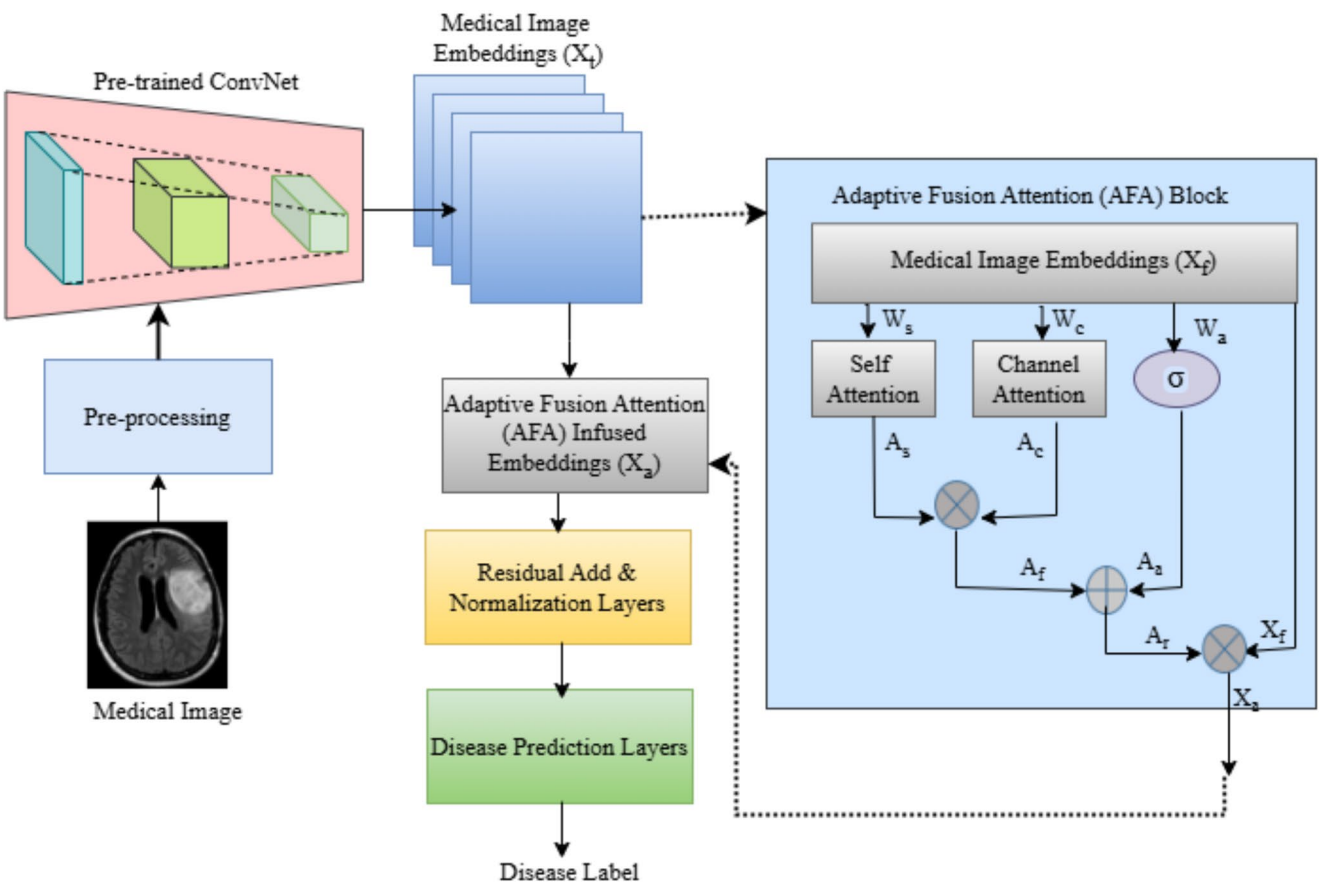

Fig. 1 Architecture of proposed Adaptive Fusion Attention Network for medical image analysis

model performance. For this study, no additional data augmentation techniques are applied, keeping the dataset static to avoid introducing synthetic variations.

3.2 Pre-trained ConvNets: learning image embeddings

After preprocessing, the next stage in the model pipeline involves extracting high-level feature representations from the medical images. This stage is crucial for transforming raw image data into abstract representations that capture complex visual patterns benefiting the subsequent stages such as classification, segmentation, or detection.

This approach leverages the EfficientNetV2B0 model, a widely adopted pre-trained deep CNN model optimized for efficiency and performance and was originally trained on the large and diverse ImageNet dataset. The standard EfficientNetV2B0 model is modified by removing its classification layers and retains only its convolutional and pooling layers. This enables the model to capture essential visual features such as edges, texture and other fundamental visual patterns relevant for medical imaging. The input medical images are passed through the modified EfficientNetV2B0 and undergoes a series of operations, including convolutional filtering, pooling, and activation functions. These series of operations enable the model to learn a rich set of high-level feature representations, suitable for downstream tasks of medical image analysis. Using this pre-trained network facilitates the extraction of relevant image features and reduces the computational cost of training the model from scratch.

The feature learning process begins by feeding the input image X_I into the modified EfficientNetV2B0 model. Mathematically, the input image is represented as a tensor $X_I \in R^{H \times W \times C}$, where H is the height of the image, W is the width, and C is the number of channels. X_I is the tensor that represents a medical image after pre-processing, just before being passed into the EfficientNetV2B0 model for feature learning. As the input tensor progresses through these layers, its spatial dimensions ($H \times W$) reduce due to operations such as pooling and striding. Simultaneously, the depth of the feature map increases, enabling the model to detect increasingly complex patterns and hierarchical features. The EfficientNetV2B0 model transforms the input tensor X_I and produces a 3D feature tensor, denoted $X_f \in R^{H^k \times W^k \times D^k}$. Here, H^k and W^k, represent the spatial dimensions of the feature map (typically smaller than the input dimensions), and D^k is the depth, indicating the number of feature channels output by the model (varies depending on the model configuration).

Pre-trained models like EfficientNetV2B0, initially trained on the ImageNet dataset, have learned general features useful across various domains. Fine-tuning allows the model to retain these features while adapting the higher layers to capture task-specific patterns, which is crucial for domain-specific data, such as medical imaging. This work proposes fine-tuning the feature learning base taken from EfficientNetV2B0 alongside other modules of the approach, including the attention module and classification head. This strategy enables the model to effectively learn task-specific information, reduce overfitting, improve generalization, and enhance performance, especially on smaller datasets like those used in medical imaging.

3.3 Adaptive Fusion Attention block

Medical images, such as MRI, CT, and retinal scans, often contain extensive information, but only specific regions typically reveal signs of disease like tumors, lesions or other pathological markers. The AFA module is designed to prioritize these critical areas and guides the model to focus on regions with diagnostically significant patterns while minimizing the impact of irrelevant details. Inspired by human visual attention, AFA enables the model to selectively focus on image regions and features most likely to carry diagnostic significance.

The AFA mechanism introduced in this work addresses the limitations of traditional spatial and channel attention mechanisms by capturing both spatial and channel-wise dependencies in a more adaptive way. Conventional attention mechanisms either focus solely on spatial features (ignoring channels) or on channel features (neglecting spatial relationships), which can lead to suboptimal performance, especially in complex real-time tasks like medical image analysis. The AFA mechanism mimics the human visual focus, where attention is directed toward areas that are most relevant, such as abnormal features in medical images (e.g., tumors or lesions), while non-essential regions are suppressed. This is achieved by combining both spatial attention (to determine "where" important features are located) and channel attention (to determine "what" features are critical for the task). In medical image tasks such as tumor detection or Covid-19 prediction, ability of the AFA module to focus on relevant areas with greater precision, while downplaying irrelevant regions, enhances the interpretability and accuracy of the model. This dynamic approach ensures that the model “looks” at the image similarly to how a doctor would-zooming in on areas of concern and leading to more accurate diagnoses and predictions. By improving the way the model attends to both spatial and channel features simultaneously, AFA presents a more effective attention mechanism for complex medical imaging

tasks. It enhances both the interpretability of the model and its performance, making it a promising addition to the field of medical image analysis.

3.3.1 Spatial attention component:

The Spatial Attention (SA) component is designed to help the model focus on critical spatial regions in medical images that are likely to contain pathologic signs, such as tumors or organ boundaries. This enables the model to prioritize potentially abnormal regions, thus enhancing its ability to make accurate predictions. The spatial attention mechanism isolates important spatial regions by utilizing a convolutional layer followed by a non-linear activation function. This structure ensures that the network captures complex spatial patterns, which are essential for tasks in medical imaging tasks such as tumor detection or organ boundary identification.

Assume that for an input image, X_I, a feature tensor X_f is obtained from a pre-trained Convolutional Neural Network (CNN) model whose dimensions are represented by $H^k \times W^k \times D^k$ where H^k is the height of the feature map, W^k is the width, and D^k represents the number of feature maps. A convolution operation is applied to this tensorX_f followed by a non-linear activation function *f*, such as ReLU or Leaky ReLU. This process is mathematically represented as:

$$F_s = f(W_s * X_f + b_s) \tag{1}$$

where X_f is the input tensor, W_s and b_s are the weights and biases of the convolution layer, $*$ denotes the convolution operation, *f* is the non-linear activation function (e.g., ReLU, Leaky ReLU). The result of this convolution is a transformed feature tensor, F_s^a, with dimensions $H^s \times W^s \times D^s$, which contains relevant spatial information. This transformation allows the model to focus on relevant spatial regions by adjusting the weights based on spatial context, effectively "highlighting" pathological regions of the medical image while minimizing the focus on irrelevant ones.

After the convolution operation, a softmax operation normalizes the spatial attention map across the spatial dimensions. The spatial attention map A_s, is obtained by applying the softmax function along each spatial axis:

$$A_s(i,j) = \frac{exp^{F_s^a(i,j)}}{\sum_l^{H^s} \sum_m^{W^s} exp^{F_s^a(l.m)}} \tag{2}$$

Here, $A_s(i,j) \in R^{H^s \times W^s}$ represents the spatial attention score at position (*i*,*j*), $F_s^a(i,j)$ is the feature map at position (*i*,*j*). This ensures that the model assigns greater importance to the most relevant regions of the image, while diminishing the influence of less important or irrelevant areas.

In the spatial attention module, the convolution layer plays a vital role by reducing the spatial dimensions of the input, which is crucial for handling high-resolution images. This dimensionality reduction results in a more compact representation that retains key spatial features. This feature map F_s is crucial for calculating attention weights in later stages. The activation function (e.g., ReLU or Leaky ReLU) enhances the representational capacity of the model, enabling it to capture non-linear relationships and complex patterns in the data. Finally, the softmax layer converts the logits into a probability distribution, enabling the model to focus on specific spatial regions. By applying the softmax function after activation, the spatial attention map $A_s^a(i,j)$ is generated, highlighting the regions that are most relevant for diagnosis. This spatial attention mechanism is particularly valuable in medical image analysis, where the understanding of spatial context is essential for accurate predictions.

3.3.2 Channel attention component

The Channel Attention (CA) component s designed to focus on prioritizing the most informative feature channels in medical images, enabling the model to focus on the channels most relevant for identifying pathological signs like tumors or organ abnormalities. By emphasizing these critical channels, the model enhances its predictive capabilities, leading to improved diagnostic accuracy.

The channel attention mechanism applies a $1\ times1$ convolutional layer to the feature tensor, X_f which is extracted from a pre-trained convolutional neural network (CNN). The use of a 1×1 convolution operation is essential for capturing channel-wise dependencies, allowing the network to weigh the significance of each channel based on its contribution to the overall feature representation. The channel attention feature map F_c is computed as follows:

$$F_c = f(W_c * X_f + b_c) \tag{3}$$

where $X \in R^{B \times H \times W \times C}$ is the input tensor, W_c and b_c are the weights and biases of the 1×1 convolution layer, *f* is the non-linear activation function such as ReLU or Leaky ReLU, and $*$ indicates the convolution operation.

Once the channel features F_c are obtained, the next step is to generate the channel attention map by applying the softmax function across the channel dimension. The channel attention scores are computed as:

$$A_c(k) = \frac{exp^{F_c(k)}}{\sum_m exp^{F_c(m)}} \tag{4}$$

where $A_c(k)$ denotes the channel attention score for channel *k*, normalized to create a probability distribution across the channel dimension. This normalization ensures that the model focuses more on the channels that carry critical information for identifying pathological features, while minimizing the influence of less informative channels.

The 1×1 convolutional layer aggregates information across the spatial dimensions, facilitating the identification of important channel-wise patterns. By condensing the feature maps into a more compact representation, this layer retains crucial information while reducing noise from less relevant channels. The activation function applied after convolution enhances the ability of the model to learn non-linear relationships, thereby improving its capacity to differentiate between critical and non-critical features.

The softmax function applied to the channel feature map results in the channel attention map $A_c(k)$, which highlights the importance of different channels in medical image analysis. This channel attention mechanism is valuable in medical applications, where certain channels may contain more significant information for diagnosis, enhancing both model interpretability and performance.

3.4 Adaptive Fusion Attention

Adaptive Fusion Attention involves the hierarchical fusion of spatial and channel attention maps, enabling the model to focus on both spatially significant regions and important channels. This fusion results in a unified attention score that reflects the relevance of each region and channel in each frame. The spatial and channel attention scores are combined through element-wise multiplication as follows:

$$A_f(i,j,k) = A_s(i,j) \odot A_t(k) \tag{5}$$

where, $A_f(i,j,k)$ represents the fused attention score for each spatial location (i,j) in frame k, emphasizing regions that are relevant both spatially and channel-wise.

This fused attention map is then applied to the input feature map, X_f:

$$X_a(i,j,k) = A_f(i,j,k) \odot X(i,j,k) \tag{6}$$

The resulting attention-weighted features, X_a emphasize critical regions over time, helping the model focus more on medically significant features across the entire sequence.

3.4.1 Adaptive attention refinement

To further refine the attention map, an additional 1×1 convolutional layer is applied, producing adaptive attention weights. A sigmoid activation is then applied to squash the output into the range [0, 1]. The adaptive attention map A_a is given by:

$$A_a = \sigma(W_a * X_f + b_a) \tag{7}$$

where σ is the sigmoid activation function:

$$\sigma(z_i) = \frac{1}{1 + exp(z_i)} \tag{8}$$

The final refined attention map A_r is obtained by adding the adaptive attention map A_a to the fused attention map A_f:

$$A_r = A_a + A_f \tag{9}$$

The refined attention map A_r have the same dimensions as A_f. The final attention-infused features can be computed by multiplying the refined attention map with the original input tensor:

$$X_a = A_r \odot X_f \tag{10}$$

This step ensures that the model attends to the most relevant regions and time steps, improving the accuracy of feature extraction and predictions.

3.5 Disease prediction module

The Disease Prediction Module is designed to classify the presence of a disease based on the attention-enhanced features extracted from the input medical images. This module processes the refined features obtained from the attention mechanism, which has emphasized the most important spatial and channel feature maps, to produce the final disease classification. The classification head consists of several layers, each contributing to the aggregation and classification of these features effectively.

The classification process includes with the Global Average Pooling (GAP) layer, which condenses the spatial information from each channel into a single value. The GAP layer computes the average of the spatial dimensions (height and width) for each feature channel, transforming the feature maps into a compact vector representation. This operation ensures that the model focuses on the presence of certain features rather than their specific locations within the image. The GAP operation can be mathematically expressed as:

$$GAP(f) = \frac{1}{H'} \frac{1}{W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} X_a(i,j,f) \tag{11}$$

where $X_a(i, j, f)$ is the attention-enhanced feature at spatial position (i, j) in feature map f, H' and W' are the height and width of the feature map X_a, respectively, and f represents the feature channels.

The GAP operation produces a summary representation for each channel, reducing the feature map to a fixed-size vector that retains only the global feature information. This reduces computational complexity and allows the model to focus on relevant patterns across the entire image. A Dense layer with ReLU activation further introduces non-linearity and captures intricate patterns in the data, helping the model refine its feature set for class discrimination. Finally, an Output layer with softmax activation converts this refined feature vector into a probability distribution across the target classes, allowing the network to output the final classification by identifying the most probable class. Together, these layers convert the attention-enhanced features into a clear and confident class prediction, enabling accurate decision-making based on the learned representations.

4 Experimental studies and discussion

This section provides a comprehensive overview of the experimental results for the proposed AFA approach across multiple disease classification tasks. It also provides an overview of experimental setup used for validation, including data preprocessing, model configuration, evaluation metrics, and a discussion on experimental outcomes.

4.1 Summary of the datasets

The proposed AFA approach has been validated across three distinct disease prediction tasks: brain tumor classification, diabetic retinopathy recognition, and COVID-19 severity prediction. Datasets for each task were chosen to provide realistic benchmarks and ensure generalizability across diverse clinical scenarios. For diabetic retinopathy recognition, the APTOS 2019 dataset [31] is utilized due to its comprehensive labeling, diverse image quality, and wide applicability in real-world clinical environments, making it ideal for model training, validation, and testing. For brain tumor classification, the Figshare dataset [23] is employed as it supports multi-class classification across different tumor types, offering a robust basis for assessing model performance in identifying varied tumor types. Finally, the SARS-CoV-2 dataset [9] is used for COVID-19 diagnosis, providing a diverse set of CT images that allows for reliable model training and testing in COVID-19 severity prediction tasks. The following subsections give a detailed overview of these datasets. Summary of the benchmark datasets is presented in Table 1 and class-wise image statistics is provided in Table 2.

Table 1 Summary of the benchmark datasets used for the evaluation of the proposed model for disease prediction tasks

Dataset	Task	Number of samples	Image dimensions	#Classes
APTOS 2019	Diabetic Retinopathy	3662 retinal images	2416 × 1736 × 3 to 819 × 614 × 3	5
Figshare	Brain Tumor Classification	3064 MRI slices	512 × 512 × 3	3
SARS-CoV-2	COVID-19 Severity Prediction	2482 CT scan images	Range from 1014 × 119 to 416 × 512 × 3	2

Table 2 Class distributions of the datasets used for disease prediction tasks

Dataset	Class name	Number of samples
APTOS 2019	No DR (0)	1805
	Mild NPDR (1)	370
	Moderate NPDR (2)	999
	Severe NPDR (3)	193
	PDR (4)	294
Figshare	Meningioma	708
	Glioma	1426
	Pituitary	930
SARS-CoV-2	COVID-19 Positive	1252
	Non-COVID-19	1230

4.1.1 APTOS-2019

The APTOS 2019 Blindness Detection dataset is a publicly available collection of 3,662 retinal images, designed for diabetic retinopathy (DR) detection and severity grading [31]. The dataset is collected from people of rural regions in India, it includes images with varying artifacts, blurriness, and exposure levels. Each image has been labeled by trained ophthalmologists according to the International Clinical Diabetic Retinopathy Disease Severity Scale (ICDRSS), categorizing the samples into five DR severity classes: no DR (49.3%), mild DR (10.1%), moderate DR (27.3%), severe DR (5.3%), and proliferative DR (8.1%). Common dimensions of the images include 2416 × 1736 × 3, 1050 × 1050 × 3, and 819 × 614 × 3.

4.1.2 Figshare

The Figshare dataset comprises brain MRI scans from 233 patients, encompassing three tumor types-Meningioma, Glioma, and Pituitary-and includes three plane views: Axial, Coronal, and Sagittal [23]. Each brain MRI scan in the

dataset has dimensions of $512 \times 512 \times 3$. The dataset contains a total of 3064 T1-weighted contrast-enhanced images, divided by tumor type into Meningioma (708 slices), Glioma (1426 slices), and Pituitary tumor (930 slices). Indices for 5-fold cross-validation are provided, enabling robust model training and evaluation. This dataset is beneficial for developing and testing models for brain tumor classification and detection in MRI scans, facilitating advancements in automated medical image analysis.

4.1.3 SARS-CoV-2

The SARS-CoV-2 CT scan is a publicly available SARS-CoV-2dataset, which includes 2482 CT images: 1252 images are from patients positive for SARS-CoV-2 (COVID-19), while the other 1230 images are from patients who are not SARS-CoV-2 positive but have other pulmonary disorders [9]. The samples are collected 120 patients (including both male and female) in hospitals across Sao Paulo, Brazil. These CT scan images are available in PNG format, with spatial dimensions ranging from 104×119 to 416×512 pixels. This publicly accessible dataset offers a valuable resource for advancing artificial intelligence (AI)-driven approaches in medical diagnostics, especially developing methods aimed at diagnosing SARS-CoV-2 infection through CT scan analysis. Using an explainable deep learning approach (xDNN), a baseline F1 score of 97.31% has been achieved on this dataset, highlighting its potential for supporting effective AI-based COVID-19 diagnostic tools.

Each dataset in this study provides a solid foundation for testing and validating the proposed AFA model across different medical image classification tasks, supporting further advancements in automated healthcare diagnostics.

4.2 Experimental setup

The experimental setup was meticulously designed to validate the performance of the proposed AFA approach across multiple disease prediction tasks, including brain tumor classification, diabetic retinopathy (DR) recognition, and COVID-19 severity prediction. The proposed approach has been implemented TensorFlow and Keras python libraries, and trained on an NVIDIA A100 GPU resulting in significant reduction in computation time and allowing for more extensive experimentation. Grad-CAM++ is employed to generate heatmaps for visualizing model attention, offering insights into how the model focuses on critical image regions for accurate disease prediction.

All the images of each dataset are resized to a fixed dimension of $224 \times 224 \times 3$ for uniformity across datasets. Cross-validation strategies used to evaluate the results are tailored to each dataset, following widely adopted approaches in the literature to ensure consistency and fairness in comparisons with existing studies. For the brain tumor classification task using Figshare dataset, which does not have predefined train-test splits, a five-fold cross-validation approach is applied, as recommended in [17]. In the case of diabetic retinopathy (DR) recognition, the APTOS-2019 dataset is divided using an 80–20 split, where 80% of the data is allocated for training and validation, while the remaining 20% is reserved for testing [21]. For COVID-19 severity prediction, the SARS-CoV-2 dataset is utilized with its predefined train-test splits, ensuring consistency with other studies employing this dataset.

4.3 Hyper-parameters

The hyperparameters for the proposed model are carefully chosen to ensure effective performance and reproducibility. These values are derived from empirical analysis and cross-validation strategies, ensuring that the model is optimized for t he task at hand. In terms of architecture, the base model used is EfficientNetV2B0, which is pre-trained on ImageNet weights. This pre-trained model serves as the feature extractor, and during training, it is fine-tuned to improve its performance on the specific task of brain cancer diagnosis. To optimize training efficiency, a batch size of 64 is selected. This batch size strikes a balance between memory utilization and model convergence speed, allowing the model to learn effectively without overloading system resources. The model is trained for 100 epochs, with early stopping incorporated to prevent overfitting. A patience of 10 epochs is applied, meaning that if there is no improvement in validation accuracy for 10 consecutive epochs, the training will be stopped early, preventing overfitting and saving computational resources.

For the optimizer, a learning rate of 0.001 is chosen for the Adam optimizer. This learning rate is ideal because Adam adapts the learning rate for each parameter, ensuring faster and more stable convergence. The categorical cross-entropy loss function is used to measure the difference between predicted and true class labels, making it suitable for this multi-class classification problem. To further enhance the training process, several callbacks are employed. Model checkpointing is used to save the best model weights during training, ensuring that the optimal model is preserved. A learning rate reduction on plateau callback is included to adjust the learning rate when validation performance stagnates, helping the model escape local minima and improving overall training efficiency. These hyperparameters and training strategies together ensure that the model is robust, efficient, and capable of high performance on the brain cancer diagnosis task.

4.4 Performance evaluation

The proposed model performance is evaluated using key metrics such as accuracy, precision, recall, F1-score, ROC-AUC, confusion matrix, and log-loss. High model accuracy indicates that the model effectively identifies both healthy and affected cases. As the medical image datasets under consideration are with high imbalance, metrics like precision, recall, F1-score are also used as they are more suitable for such imbalance datasets. High precision reduces the risk of false positives, indicating fewer healthy cases are misclassified as diseased. High recall in medical fields is crucial because failing to detect a disease (false negatives) can have serious consequences. The F1-score balances precision and recall, providing an overall measure of model performance. In addition, ROC-AUC is also used evaluates the ability of the model to distinguish between classes (e.g., cancerous vs. non-cancerous). A high AUC means the model is very good at distinguishing the diseases. Log-loss measures how well the predicted probabilities align with actual labels, penalizing the model for incorrect confident predictions, and is used during training to optimize performance.

4.5 Experimental outcomes and discussion

The proposed AFA model has been evaluated on three disease prediction tasks-brain tumor classification (BTC), diabetic retinopathy (DR) recognition, and COVID-19 severity prediction. Table 3 summarizes the results obtained by the proposed approach on these benchmark datasets in terms of accuracy, precision, recall, F1-score and AUC.

4.5.1 Results on APTOS 2019 dataset for diabetic retinopathy severity classification

For the task of Diabetic Retinopathy (DR) Severity Classification using the APTOS 2019 dataset, the AFA model achieves an accuracy of 83.55%, with a macro precision of 84.22% and an ROC AUC score of 95.34%. These metrics highlight robust discriminative power of the model, showing its ability to effectively identify varying levels of DR severity.

The results from Table 4 highlight both the strengths and limitations of the AFANet model in classifying diabetic retinopathy (DR) stages. The model performs exceptionally well in identifying the "No DR" class, with high precision (98.39%) and recall (98.92%). This indicates that the model is very effective in distinguishing images without diabetic retinopathy, which is likely due to the abundance of such cases in the dataset and the relatively simple features of this class.

Table 3 Performance of the proposed Adaptive Fusion Attention model on different disease prediction datasets

Dataset	Accuracy	Precision	Recall	F1-Score	ROC AUC
Figshare	98.69	98.33	98.83	98.57	99.94
APTOS 2019	84.56	84.22	83.49	83.61	95.34
SARS-CoV-2	99.80	99.80	99.80	99.80	99.98

Table 4 Disease specific scores on the test split of the APTOS 2019 dataset using proposed AFANet approach

Class	Precision	Recall	F1-Score	Support
No DR	98.39	98.92	98.66	371
Mild	68.85	60.00	64.12	70
Moderate	72.29	87.43	79.15	191
Severe	52.00	33.33	40.63	39
Proliferate	71.43	49.18	58.25	61
Accuracy	84.56 (732)			
Macro avg	72.59	65.77	68.16	732
Weighted avg	84.04	84.56	83.80	732

However, the performance declines for more severe DR stages. The "Mild" class shows moderate precision (68.85%) but low recall (60.00%), suggesting that while the model correctly identifies some mild cases, it fails to identify many others. This could be due to the subtle differences in features between "No DR" and "Mild" stages, making it harder for the model to differentiate them accurately. Similarly, the "Moderate" class shows improved recall (87.43%) but a lower precision (72.29%), indicating that the model is able to detect most moderate cases but also misclassifies some as "No DR" or "Severe."

The "Severe" and "Proliferate" classes exhibit significant challenges. The model struggles with low recall values (33.33% and 49.18%, respectively), which may be due to the relatively fewer examples of these severe stages in the dataset. The imbalance in class distribution, with more "No DR" and "Moderate" cases, could contribute to this difficulty in detecting severe stages. Additionally, the fine-grained visual differences between severe DR stages and other classes might make it harder for the model to learn accurate representations for these classes.

The macro average results (72.59% precision and 65.77% recall) reflect the model's overall moderate performance, while the weighted average (84.04% precision and 84.56% recall) indicates that the model's performance is heavily influenced by the dominant "No DR" class. These results suggest that while the model is quite effective for detecting the absence of DR, there is a need for further refinement, and improving detection of severe stages.

The confusion matrix shown in Fig. 2 reveals that the model performs well in classifying "No DR," with 99.1% of the samples correctly classified. However, it struggles with the "Mild" and "Proliferate" classes, misclassifying a significant portion of these as "Moderate" and "Severe."

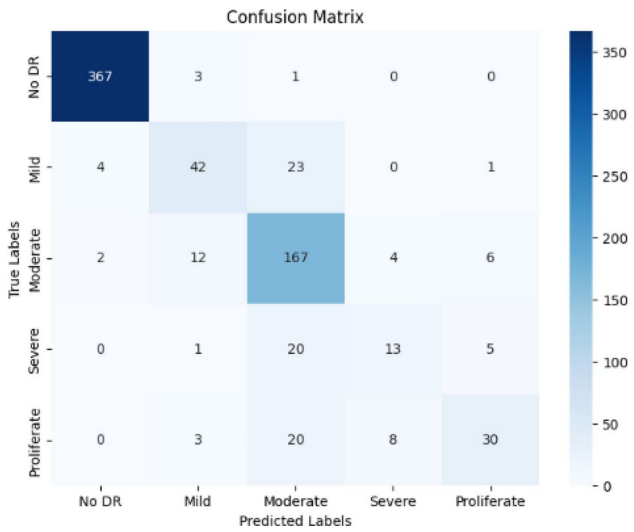

Fig. 2 Confusion matrix for the test split of APTOS-19 dataset using the proposed AFANet model for diabetic retinopathy severity grade classification

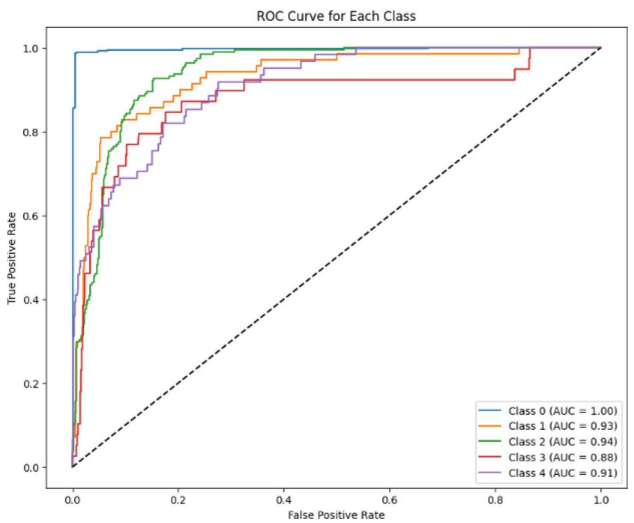

Fig. 3 ROC curve of proposed AFANet for diabetic retinopathy severity grade classification

Specifically, 32.9% of "Mild" samples were misclassified as "Moderate," and 51.3% of "Severe" samples were misclassified as "Moderate." These errors suggest that the model faces challenges in differentiating between certain classes, likely due to similarities in their visual characteristics. This highlights the need for more refined models or additional training data to improve performance on these more ambiguous categories.

The AUC curves presented in the Fig. 3 demonstrate excellent performance across all classes, with the "No DR" class achieving the highest AUC score of 99.60. The micro-average AUC of 96.70 reflects strong overall classification performance, while the macro-average AUC of 93.00 indicates a more balanced performance across all classes, despite some challenges in distinguishing between more complex categories like "Severe" and "Proliferate".

For diabetic retinopathy detection, the AFA mechanism enhanced ability of the model to identify subtle lesions in retinal images. By highlighting both the spatial regions associated with abnormalities and the specific channels carrying critical features, the model improved its diagnostic accuracy, especially for early-stage retinopathy where abnormalities are less pronounced.

Table 5 Classification report for Figshare dataset (class-wise scores)

Class	Precision	Recall	F1-score	Support
Meningioma	97.92	99.30	98.60	142
Glioma	100.00	98.95	99.47	285
Pituitary	99.47	100.00	99.73	186
Accuracy	99.35 (613)			
Macro Avg	99.13	99.41	99.27	613
Weighted Avg	99.36	99.35	99.35	613

4.5.2 Results on FigShare dataset for brain tumor classification

As shown in Table 3, for Brain Tumor Classification (BTC), the AFANet model achieves an accuracy of 98.69% and an ROC AUC score of 99.94%. The macro F1-Score of the model is 98.57% reflects balanced performance across the three tumor classes (Meningioma, Glioma, and Pituitary), demonstrating reliable classification.

Table 5 presents the performance of the proposed AFANet model on Figshare dataset for Brain Tumor Classification (BTC) task. The scores in table demonstrate that the AFANet model achieves exceptional performance by achieving a high overall accuracy of 98.69% and an ROC AUC score of 99.94%, indicating excellent predictive capability. This performance is balanced across the three tumor classes: Meningioma, Glioma, and Pituitary, with a macro F1-score of 98.57%, reflecting robust results across classes without significant bias toward any specific tumor type. Class-specific metrics reveal that the model is precise and consistent across all tumor types. For Meningioma, the model achieves a precision of 97.92%, a recall of 99.30%, and an F1-score of 98.60%, highlighting its reliability in identifying Meningioma cases. In Glioma classification, AFANet attains a precision of 100.00%, a recall of 98.95%, and an F1-score of 99.47%, showing both high accuracy and a strong ability to correctly classify Glioma instances. For Pituitary tumors, the model achieves a precision of 99.47% and a perfect recall of 100.00%, resulting in an F1-score of 99.73%, which underscores the model is consistent in identifying Pituitary cases accurately. These results confirm that the proposed approach is reliable and highly capable in distinguishing between different brain tumor types,

providing a valuable tool for medical image analysis and tumor diagnosis.

Examining the confusion matrix shown in Fig. 4, AFANet misclassified only 3 out of 613 samples of the test split, showing low error rates for each class. The classification report provides high precision, recall, and F1-scores across classes, indicating consistent, accurate performance without significant variance.

Figure 5, presents the AUC scores for each class in the Figshare dataset. The AUC scores infer the excellent performance by the AFANet model, with Meningioma, Glioma, and Pituitary achieving AUC scores of 99.73, 99.83, and 99.99, respectively. These high scores indicate that the model is strong in distinguishing between the tumor types. The micro-average AUC of 99.92, calculated by considering all classes equally, confirms outstanding overall performance, while the macro-average AUC of 99.85, which treats each class equally, also reflects balanced performance across all tumor types. Overall, the results suggest that the model is highly effective in classifying the tumor types, with particularly strong performance in the Pituitary class.

In the case of tumor detection, the AFA model is able to identify tumor regions with greater precision, significantly reducing false positives and false negatives. By dynamically focusing on both spatial areas and critical channels, the model achieved higher sensitivity and specificity, leading to more accurate predictions of tumor presence.

Figure 6 depicts how the model is improving accuracy and reducing the loss over epochs. The model exhibits excellent performance on the Figshare dataset, achieving a maximum training accuracy of 99.00 and a validation accuracy of 98.00 after 50 epochs, indicating strong generalization and minimal over-fitting. The training loss was 0.02, and the validation loss was 0.04, both of which are low, reflecting ability of the model to accurately predict tumor types with minimal error. The slight difference between training and validation accuracy, as well as between training and validation loss, suggests that the model effectively learned the essential features without over-fitting, achieving a balance between learning from the data and generalizing well to unseen samples.

4.6 Result analysis for COVID-19 diagnosis

For COVID-19 Severity Prediction, the model demonstrates an exceptional capability, achieving an accuracy of 99.80% and an ROC AUC score of 99.98%. These scores reflect precise identification and differentiation of COVID-19 severity levels, suggesting that the model is practically applicable in a clinical environment.

The classification report presented in Table 6 provides a detailed evaluation of the proposed model performance on

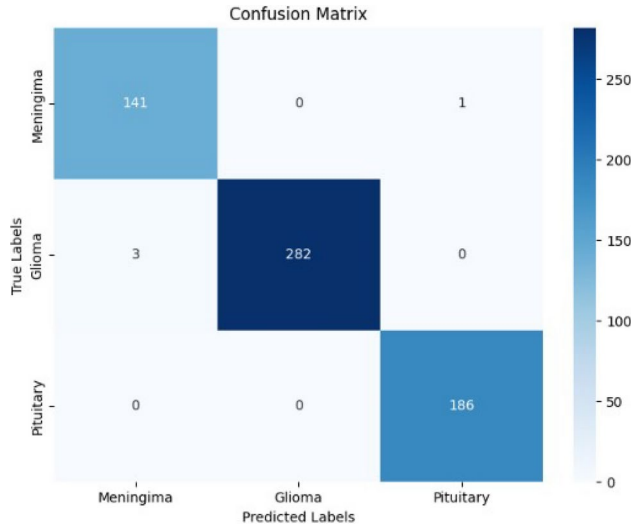

Fig. 4 Confusion Matrix for the test split of the Figshare dataset using the proposed AFANet model for brain tumor recognition

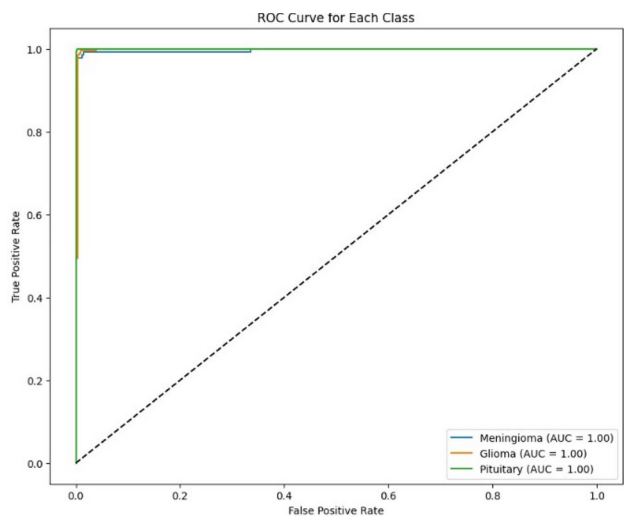

Fig. 5 ROC Curve of proposed AFANet for tumor type recognition

the dataset. The table lists the precision, recall, and F1-score for both covid and non covid cases. For both classes, the precision, recall, and F1-score are 99.60% and 99.59%, indicating that the model performs very well in terms of accurately predicting the covid while minimizing false positives and false negatives. The overall accuracy of the model is 99.60%, suggesting that it correctly classifies almost all of the test samples. The macro average and weighted average scores, which account for both the individual class performance and the overall class distribution, also show excellent results, with all values consistently at 99.60%. This indicates that the model is balanced in its performance across both cases of covid, and it generalizes well without bias toward any particular class.

Figure 7 depicts how the model is improving accuracy and reducing the loss over epochs. As the model trains, the

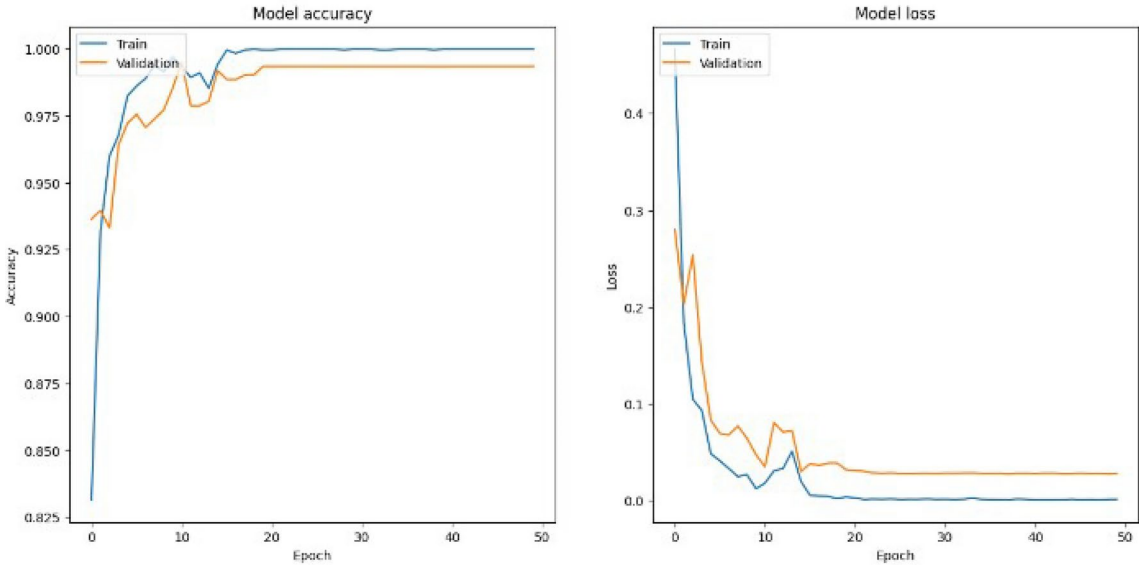

Fig. 6 Visualization of training and validation accuracy, loss over epochs for the figshare dataset Using AFANet

Table 6 Classification report for SARS CoV-2 dataset (class-wise scores)

Class	Precision	Recall	F1-score	Support
Covid	99.60	99.60	99.60	251
Non Covid	99.59	99.59	99.59	246
Accuracy	99.60 (497)			
Macro avg	99.60	99.60	99.60	497
Weighted avg	99.60	99.60	99.60	497

training accuracy steadily increases, indicating that it is successfully learning from the data. Similarly, the validation accuracy increases and eventually plateaus, suggesting that the model is generalizing well to unseen data. The plateau in validation accuracy signifies that the model has reached a point where further performance improvements are minimal, indicating it has effectively learned the underlying patterns. This behavior is common in well-optimized models, where both training and validation accuracies converge after several epochs. The same trend is seen in the loss curves. As the model trains, the training loss steadily decreases, showing improved predictions and reduced error on the training data. The validation loss also decreases and then plateaus, signaling that the model is not overfitting and continues to generalize well. This plateau in validation loss

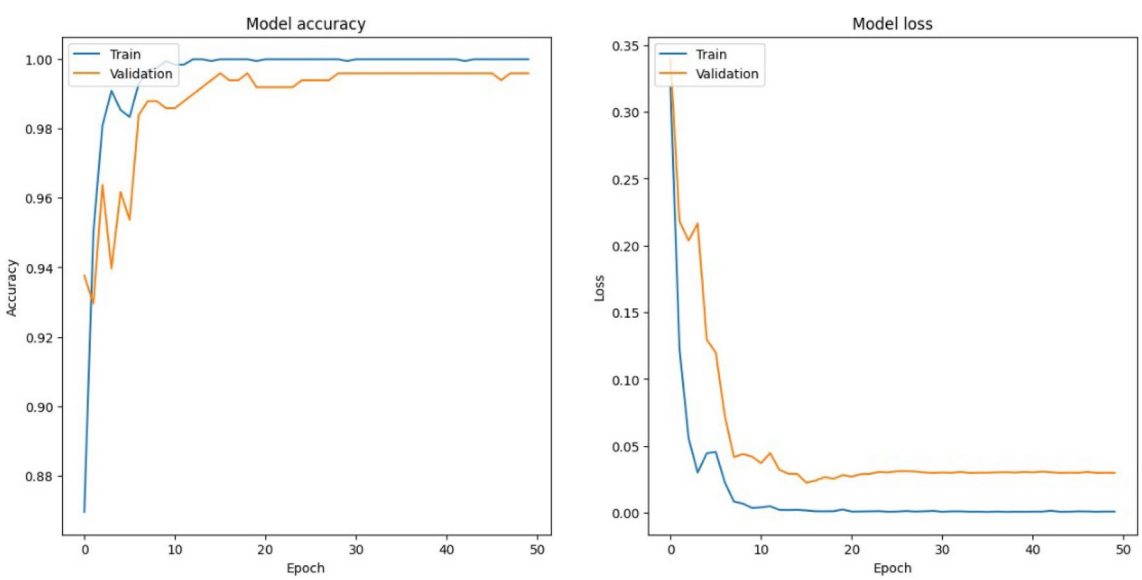

Fig. 7 Visualization of training and validation accuracy, loss over epochs for the SARS CoV-2 dataset using AFANet

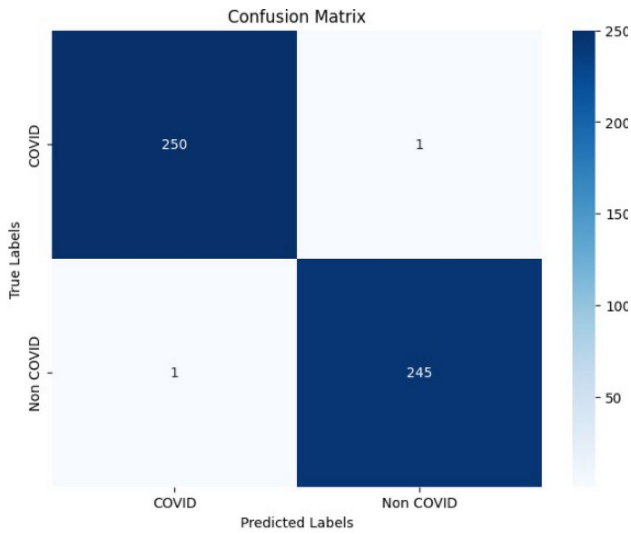

Fig. 8 Confusion matrix of proposed AFANet for COVID-19 diagnosis

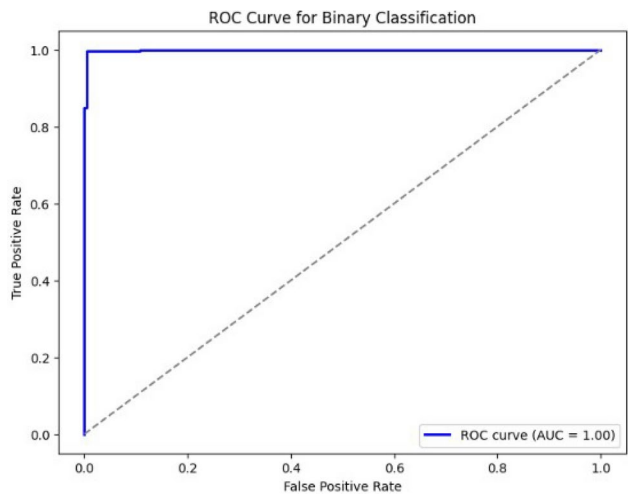

Fig. 9 ROC curve of proposed AFANet for COVID-19 diagnosis

further confirms that the model has reached an optimal state, with consistent error rates on new data, reflecting its ability to minimize both training and validation losses effectively.

Figure 8 presents the confusion matrix for COVID and non-COVID cases and shows that the model performs well with 250 true negatives and 245 true positives, indicating accurate identification of both classes. There is only 1 false positive and 1 false negative, which highlights ability of the model to minimize errors in classification. This results in a high accuracy of 99.60%, demonstrating strong performance with very few mis-classifications of CT scan images of Covid patients.

Figure 9 represents the ROC Curve illustrating category specific area under curve values. AUC for both Covid and Non-Covid class is almost 100% with 0.02% difference. The AUC scores for the COVID and non-COVID classes are all 0.9959754478022869, which indicates excellent model performance in distinguishing between the two categories. The high AUC value reflects the model's strong ability to discriminate between COVID and non-COVID cases. Since both the micro-average and macro-average AUC scores are the same, it suggests that the model is performing consistently across both classes, without significant bias toward either COVID or non-COVID cases. A value close to 1.0 indicates that the model is highly effective in classifying the cases accurately.

In Covid-19 prediction tasks, the AFA mechanism improved the identification of relevant features in chest X-rays and CT scans. By leveraging both spatial and channel attention, the model effectively distinguished between normal and Covid-19-affected areas, improving its overall performance and robustness in predicting disease severity.

Table 7 Performance comparison of diabetic retinopathy (dr) recognition models on the APTOS 2019 dataset

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
Ensemble of CNNs [5]	80.41	80.21	80.61	80.52	–
VGG16 Fine-Tuned [40]	82.23	82.57	82.10	82.34	92.17
ResNeXt Model [30]	82.99	83.20	83.00	83.12	93.12
CNN512 based Lesion Localisation System [7]	84.10	–	–	–	–
Lesion Aware Attention Net [53]	84.17	–	–	–	–
Stacked CAE with SA [15]	84.17	–	–	–	–
Hinge Attention Net [54]	84.17	–	–	–	–
Multi-Scale Attention Net [6]	84.60	–	–	–	–
Patch-division based DenseNet [32]	84.90	–	–	–	–
Proposed AFANet	84.56	84.22	83.49	83.61	95.34

4.7 Comparison analysis

This section provides comparison of the proposed AFANet with recent literature. This will enable to understand the superiority of the model compared to the literature.

4.7.1 Comparison of AFANet with literature on APTOS-2019 fundus image dataset

Table 7 presents a comparison of various models for Diabetic Retinopathy (DR) recognition on the APTOS 2019

dataset, evaluating performance across accuracy, precision, recall, F1-score, and ROC AUC. The Ensemble of CNNs model shows an accuracy of 80.41%, but lacks an ROC AUC score, indicating limited performance metrics. The VGG16 Fine-Tuned model achieves an accuracy of 82.23%, with a solid ROC AUC of 92.17%, highlighting its strong ability to differentiate DR stages. The ResNeXt Model performs slightly better with an accuracy of 82.99% and a ROC AUC of 93.12%, demonstrating good balance in precision and recall.

Other models, such as the CNN512-based Lesion Localisation System and Lesion Aware Attention Net, show accuracies of 84.10% and 84.17%, respectively, but lack complete metric data. Models like Multi-Scale Attention Net (84.60% accuracy) and Patch-division based DenseNet (84.90% accuracy) also perform well, though full metric results are not provided. The Proposed AFA Model stands out with an accuracy of 83.55%, high precision and recall values (84.22% and 83.49%), and an exceptional ROC AUC of 95.34%, making it superior in the comparison for distinguishing DR stages effectively.

Table 8 Performance comparison of tumor classification models with the proposed approach on the Figshare dataset

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Capsule Network [3]	90.89	–	–	–	–
Hybrid CNN & DNN [43]	91.43	–	–	–	–
VGG-19 Fine tuning [60]	94.82	–	–	–	–
Two Channel DNN [19]	95.23	–	–	–	–
AlexNet + SVM [49]	95.77	–	–	–	–
Custom CNN + SVM [24]	95.82	–	–	–	–
TL + VGG19 [1]	95.84	95.33	95.55	95.42	96.47
3D-CNN Model [26]	96.50	96.12	96.20	96.15	97.33
Hybrid CNN-RNN [35]	97.65	97.31	97.89	97.60	98.20
MSENet [20]	96.05				
MANet [55]	96.51				
MedTransNet [58]	98.37				
AFANet (proposed)	98.69	98.33	98.83	98.57	99.94

4.7.2 FigShare MRI scan image dataset

The performance comparison reported in Table 8 showcases the progressive advancements in brain tumor classification models, moving from traditional CNN and hybrid approaches to state-of-the-art architectures with attention mechanisms and transformers. Initial models, like the Capsule Network and hybrid CNN & DNN, provided moderate accuracy, but lacked the sophistication needed to capture complex MRI features fully. Fine-tuned architectures like VGG-19 and custom CNN + SVM combinations improved accuracy and generalization by leveraging pre-trained layers or SVM classifiers, although they still fell short of models with attention capabilities.

Attention-based models, including MSENet, MANet, and MedTransNet, brought substantial improvements by focusing on the most relevant MRI regions, with MedTransNet achieving 98.37% accuracy using transformers to capture long-range dependencies. The proposed AFANet model surpasses all prior models with an impressive 98.69% accuracy and the highest values across precision, recall, F1-score, and ROC-AUC, demonstrating an exceptional ability to handle complex image features. This evolution underscores the effectiveness of attention mechanisms and transformer-based designs in advancing the accuracy and reliability of brain tumor classification.

4.7.3 CoVID-19 CT scan image dataset

Table 9 compares the performance metrics of various state-of-the-art models for COVID-19 severity prediction, as per the provided dataset. The table compares various state-of-the-art models for COVID-19 severity prediction using the SARS-CoV-2 dataset, highlighting their performance across accuracy, precision, recall, F1-score, and ROC AUC. The Proposed AFA Model outperforms all other models with exceptional results, achieving an accuracy, precision, recall, F1-score, and ROC AUC of 99.80% and 99.98%, respectively. This signifies its highly effective ability to predict COVID-19 severity. The Baseline CNN Model shows a slightly lower accuracy of 99.60%, but lacks other performance metrics, suggesting it might be limited in comprehensive evaluation.

The VGG16 with Transfer Learning model, a common choice in the literature, achieves 98.70% accuracy with strong macro precision, recall, and F1-scores. The Hybrid CNN-LSTM model performs well with a 99.10% accuracy, but with a slightly lower recall compared to the AFA model. EfficientNet-B7 achieves 99.50% accuracy, with strong precision and recall values, similar to the AFA model but with slightly lower F1-score. Other notable models, such as xDNN (97.30% accuracy) and Contrastive Learning (90.83%

Table 9 Performance comparison of various state-of-the-art models with proposed AFA model on SARS-CoV-2 dataset

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
xDNN [9]	97.30	99.10	95.50	97.30	
Contrastive Learning [65]	90.83	–	–	–	
DenseNet201 [29]	96.20	–	–	–	
Two-stage Xception + FS [14]	97.16	–	–	–	
COV-CAF [27]	97.59	–	–	–	
Modified VGG-19 [42]	95.00	95.30	94.00	94.30	
Convolutional SVM [41]	96.00	–	–	–	
Bi-stage FS + CNN [52]	95.77	–	–	–	
5 Clf Novel literature model [56]	98.99	99.02	98.97	98.99	
8 Clf Novel literature model [56]	98.99	98.98	99.00	98.99	
AFANet (Ours)	99.60	99.60	99.59	99.60	

accuracy), perform well but are less effective compared to the top-performing models in the comparison. The proposed AFANet model also achieves an impressive 99.60% accuracy, precision, and recall, demonstrating its competitive edge in predicting COVID-19 severity effectively.

After rigorous experimentation across BTC, APTOS2019 (for DR), and SARS-CoV-2 datasets, it is trivial that the proposed model achieves improved performance over existing approaches, particularly in terms of accuracy, F1-score, and ROC AUC. This validates the robustness and effectiveness of proposed attention approach across different medical image classification tasks.

4.8 Attention visualization and interpretability

To further understand the impact of the proposed AFA, several visualization studies have been conducted using attention maps to analyze the areas the model focuses on with and without the AFA module. Results indicate that models equipped with the AFA consistently attend to more relevant regions, ignoring noise and irrelevant details. The attention maps generated by the AFA exhibit clear and interpretable patterns, highlighting objects and regions that align closely with human visual perception.

This section presents visual interpretations of the proposed AFANet model performance and visualizes how the

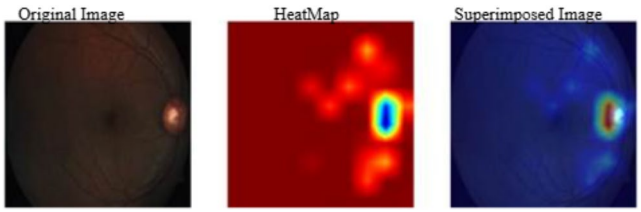

Fig. 10 No DR fundus image showing with grad-CAM overlay

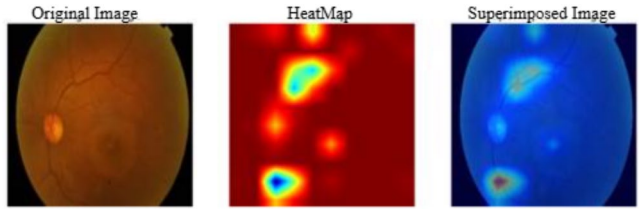

Fig. 11 Mild DR fundus image showing with grad-CAM overlay

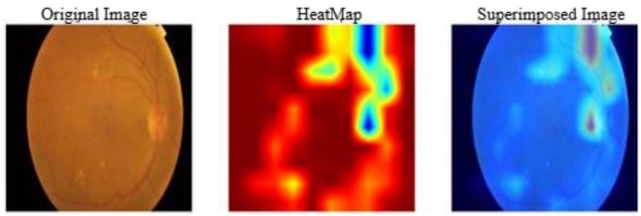

Fig. 12 Moderate DR fundus image showing with Grad-CAM overlay

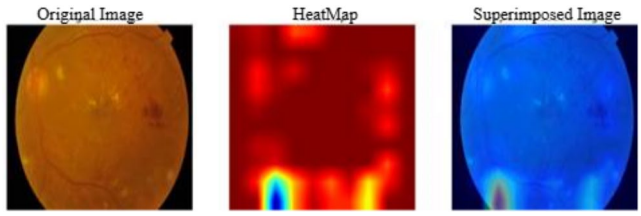

Fig. 13 Severe DR fundus image showing with Grad-CAM overlay

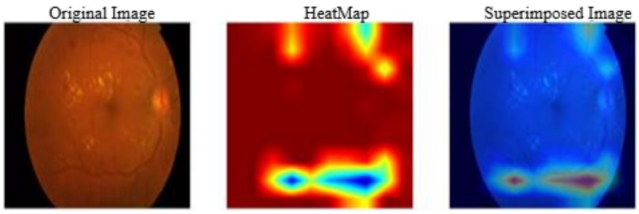

Fig. 14 Proliferate DR fundus image showing with Grad-CAM overlay

proposed dynamic context attention helps in improving the model superior to existing attention based approaches. Figures 10,11, 12, 13, and 14 shows Grad-CAM overlays on the original fundus images, highlighting the regions the model considers most relevant in its classification of diabetic retinopathy (DR) stages. In diabetic retinopathy (DR), each severity stage is characterized by specific pathological signs, which the model identifies and highlights as part of its interpretive process. The attention mechanism integrated in

the proposed model effectively targets these regions, aligning with known clinical markers. For healthy fundus images without DR, the model focuses on the optic disc, confirming an understanding of normal retinal structures. As shown in Fig. 10, no pathological markers are present in no DR case, which aligns with a clean focus on anatomical landmarks without highlighting abnormal areas. The spatial and channel attention maps demonstrate that AFA not only identifies object boundaries effectively but also captures textural and contextual details that enhance its classification accuracy.

In mild DR cases, the model primarily focuses on microaneurysms-small, round swellings of blood vessels caused by damaged capillary walls. These lesions are often the first visible signs of DR, and the model attention mechanism effectively captures them as critical early indicators as shown in Fig. 11. With moderate DR, attention is directed toward a combination of hemorrhages and exudates (lipid or protein deposits). Hemorrhages may appear as small red spots or dots, while exudates are often yellow or white. AS shown in Fig. 12, the model focuses on these regions, accurately identifying the presence of moderate damage that impacts the retina more broadly. Severe DR is marked by extensive intra-retinal hemorrhages, venous beading, and intraretinal microvascular abnormalities (IRMA)-all indicative of substantial blood vessel damage and ischemia. Attention mechanism of the model highlights these larger regions of hemorrhages and vessel irregularities, capturing areas that indicate significant disease progression as shown in Fig. 13. In the proliferative stage, new abnormal blood vessels (neovascularization) start to grow on the retinal surface as a response to ischemic damage. AS shown in the Fig. 14, dynamic attention in the model allows the model to focus on these regions of neovascularization and large hemorrhages, showing its ability to detect the severe pathological response of the retina to oxygen deprivation.

To assess the impact of the proposed AFA module, visualizations have been conducted on brain tumor classification (BTC) using attention maps to analyze the areas the model focuses on with and without AFA. In the case of BTC, the proposed AFA consistently identifies critical regions in MRI scans that correspond to the tumor's location and features. Figures 15, 16, 17, and 18 show MRI scans with Grad-CAM overlays, demonstrating how the AFA module highlights the areas most relevant for tumor detection. The attention maps effectively focus on irregularities such as abnormal growth patterns, tissue deformations, and other signs of malignancy that may be challenging to detect without an attention mechanism.

In the absence of AFA, the model tends to focus on irrelevant regions, or it may struggle to emphasize the most indicative features of tumor. However, the AFA module enhances this by providing a more refined and targeted attention,

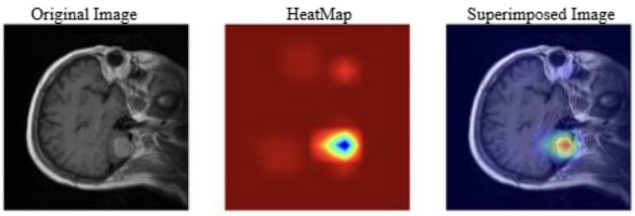

Fig. 15 MRI image showing with Grad-CAM overlay-sample 1

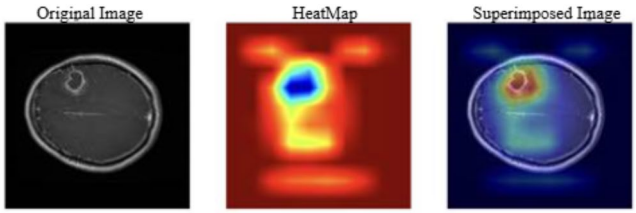

Fig. 16 MRI image showing with Grad-CAM overlay-sample 2

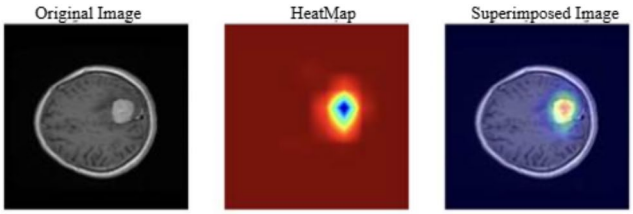

Fig. 17 MRI image showing with Grad-CAM overlay-sample 3

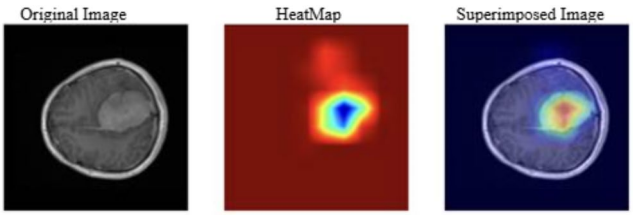

Fig. 18 MRI image showing with Grad-CAM overlay-sample 4

which improves the model accuracy in classifying different stages and types of brain tumors. The ability of the AFA to emphasize tumor types, including pituitary, meningioma, or Glioma, demonstrates its potential to help clinicians make more informed decisions during diagnostic processes.

From these visualizations it is evident that the proposed AFA mechanism enables the model to adapt its focus to the unique pathological markers associated with each stage of the disease, enhancing diagnostic accuracy and interpretability in detecting DR progression. This interpretability is valuable in real-world applications, as it provides a transparent decision-making process that can be easily validated by domain experts, such as clinicians or quality control engineers.

Overall, the inclusion of the AFA layer provides an advanced level of attention modulation, enabling enhanced

interpretability and flexibility in handling diverse medical image data. This work highlights the potential of the AFA mechanism as a valuable addition to the toolkit for medical image analysis, setting a new standard in model accuracy, reliability, and explainability in healthcare applications.

5 Conclusion and future research directions

In this study, we introduced the AFA mechanism to enhance the performance of deep learning models for medical image analysis. Traditional attention mechanisms, such as spatial and channel attention, often neglect the interdependencies between spatial and channel dimensions, limiting their effectiveness in complex tasks like pathology detection. The AFA mechanism addresses this issue by integrating both spatial and channel attention, allowing the model to dynamically focus on critical regions and relevant channels for better diagnostic accuracy.

The AFANet has been evaluated on three significant diseases: tumor detection, diabetic retinopathy, and Covid-19 prediction. In each case, the AFA mechanism demonstrated improved performance over traditional attention methods. The results from these three experiments confirm that the AFA mechanism provides a significant boost to ability of the model to focus on relevant features, leading to more precise and reliable predictions. The AFA model not only enhances accuracy but also provides greater interpretability, helping healthcare professionals better understand and trust the model predictions. In conclusion, the AFA mechanism is a promising advancement for medical image analysis, offering improved accuracy and robustness across various diseases, with potential for real-time applications in clinical settings. Future work can explore its applicability to other medical conditions, further advancing AI-driven diagnostic systems.

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